

Security Testing Report - Task 3

1. Objective

Perform security testing on DVWA by identifying common web vulnerabilities and documenting mitigation techniques.

2. Environment

- Kali Linux
- DVWA
- Apache2
- MariaDB
- PHP
- Burp Suite Community Edition

SQL Injection

Payload: ' OR '1'='1

Result: Retrieved all user records.

Mitigation: Prepared Statements, Parameterized Queries, Input Validation.

Screenshot: 02_SQL_Injection.png

Reflected XSS

Payload:

Result: JavaScript executed.

Mitigation: Input Validation, Output Encoding, CSP.

Screenshot: 03_Reflected_XSS.png

Stored XSS

Result: Stored script executed for users.

Mitigation: Output Encoding, HTML Sanitization, CSP.

Screenshot: 04_Stored_XSS.png

CSRF

Result: Password changed without CSRF protection.

Mitigation: CSRF Tokens, SameSite Cookies.

Screenshot: 05_CSRF.png

Local File Inclusion

Payload: ../../../../etc/passwd

Result: File disclosure.

Mitigation: Path validation, Allowlists.

Screenshot: 06_File_Inclusion.png

File Upload

Uploaded PHP file demonstrating insecure upload.

Mitigation: File type validation, Store uploads outside web root.

Screenshot: 07_File_Upload.png

Burp Suite Analysis

Captured requests, configured Intruder, analyzed HTTP headers. Missing headers include CSP, X-Frame-Options and HSTS.

Screenshots: 08_Burp_Intruder.png, 09_HTTP-Headers.png

Conclusion

DVWA vulnerabilities were successfully demonstrated and mitigation techniques were documented.